Describe the type(s) of research subjects or activities that interest you at your first and second choice host laboratories, and discuss any particular factors influencing your choice of host laboratories.

My research interests are centered around modeling complex, physical systems through various computational frameworks, including quantum algorithms, molecular biophysics, combinatorial optimization and machine learning. Oak Ridge National Laboratory's (ORNL) focus on quantum algorithm development, research in hybrid quantum-classical computing, and initiatives in molecular biophysics strongly align with my interests and would provide me with invaluable training for a future career in computational science.

I have worked on advancing quantum circuit simulation techniques using group theoretic methods for reducing T-count, and I am eager to explore alternative approaches to quantum algorithm design and simulation at ORNL. I'm particularly excited to work in the Quantum Computational Science Group under Dr. Ryan Bennink, whose research in lattice surgery and quantum error correction builds on my current experience and will expand my knowledge of the challenges associated with scalable quantum computing. Additionally, through my role as a research assistant in the MSU-DOE Plant Research Laboratory, I have also developed a strong passion for applying computational methods to understand biomolecular processes. ORNL's Molecular Biophysics Group offers the opportunity to use large-scale molecular dynamics simulations, deep learning structure prediction, and enhanced sampling techniques to study membrane dynamics and protein-ligand interactions in support of bioenergy research, an area I find fascinating. Furthermore, ORNL's Q-HPC initiative to develop a hybrid-quantum classical high performance computing environment integrates my interests in quantum algorithms, molecular simulation, and HPC methods.

Brookhaven National Laboratory (BNL) offers research opportunities in computational science and machine learning that would enhance my technical skills and allow me to meaningfully contribute to ongoing projects. I'm particularly excited about the prospect of working on developing quantum algorithms for simulating quantum chromodynamics and other high energy physics phenomena as part of the Computing and Data Sciences (CDS) directorate. As part of the Computational Science Initiative (CSI), BNL also provides the chance to work on exascale numerical algorithms for scientific applications in chemistry, material science, and biology, from which I can better understand what scientific problems require advanced computational resources. These programs would allow me to expand my computational experience across multiple scientific domains, which is important to me as I seek to pursue a career in computational science research.